MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL

Department of Electrical Engineering

B. Tech. III Semester Mid Examination October 2025

Sub: Measurement & Instrumentation (EE-24214)

Max. Marks: 20 **Time: 60 Minutes**

Note: Solve all questions else do as directed. Assume any missing data with due justification. Approximate to two decimal. Calculators are allowed.

		COs	Marks
Q.1	How does one extend the range of a voltmeter and an ammeter?	CO1	3
Q.2	Obtain the mathematical expression for deflecting torque and controlling torque for the DC ammeter.	CO1	3
Q.3	Why the pole faces of PMMC-type instruments are curved? Why do we have the cylindrical core?	CO2	2
Q.4	Why is it impossible to shape the flux lines in the electrodynamometer instruments?	CO2	2
Q.4	A manufacturer wants to measure the resistance of the device. However, in the warehouse of the manufacturer, only some instruments are available, such as a d'Arsonval meter with 40mA full scale deflection current and an internal resistance of 10Ω, 5Ah 3V battery, a potentiometer of 500Ω and resistors of various values. Design a shunt ohmmeter by adding appropriate shunt resistance across the movement so that the shunt ohmmeter indicates 5Ω at the mid-point on its scale. Also, find the exact value of the resistance that the potentiometer should be kept at for the ohmmeter to operate without any damage. What does ohmmeter read at full scale deflection, if the accuracy of d'Arsonval meter is 95%?	CO3	5
Q5	An electrodynamometer ammeter and a moving iron ammeter are connected in series with a 100 Ω resistor. A voltage having sine waveform and a frequency of 50 Hz and peak value of 100V is applied to the circuit. The mutual inductance of an electrodynamometer varies with deflection as $M = 6\theta + 30$ uH, whereas in a moving iron ammeter, the self-inductance varies with deflection as $L = (0.01 + 0.05\theta) \times 2$mH. Calculate the deflection of each ammeter in radians. Assume the 1.67×10^{-6}N-m/degree spring constant to be the same for both meters. What is the range of meters?	CO3	5

Date of Examination: 01/10/2025